

Below are solutions/hints on some of the questions. There is no guarantee of completeness or correctness; please watch out for typos.

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Integrals you might need:

$$\int_0^\pi du \sin^2 u = \frac{\pi}{2}, \quad \int_0^\pi du u \sin^2 u = \frac{\pi^2}{4}, \quad \int_0^\pi du u^2 \sin^2 u = \frac{\pi^3}{6} - \frac{\pi}{4}.$$

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1. A particle is confined in the one-dimensional region between $x = 0$ to $x = L$. It is in the stationary state $\psi = \sqrt{\frac{2}{L}} \sin(\pi x/L)$.

(a) [**3 pts.**] Show that the wavefunction is normalized.

Solution $\rightarrow$

$$\begin{aligned} \langle \psi | \psi \rangle &= \int_0^L dx \psi^*(x) \psi(x) = \frac{2}{L} \int_0^L dx \sin^2 \left(\frac{\pi x}{L} \right) \\ &= \frac{2}{L} \frac{L}{\pi} \int_0^\pi du \sin^2(u) \quad [\text{substitution } x = Lu/\pi] \\ &= \frac{2}{L} \frac{L}{\pi} \frac{\pi}{2} \quad [\text{using the integral given}] \\ &= 1 \quad \text{Normalized!} \end{aligned}$$

Comments about the integral:

(1) If you plot the integrand $\sin^2 u$, you can convince yourself that the integral should be $\pi/2$.

(2) When you change variables inside a definite integral, the limits of integration need to be changed as well.

- (b) [6 pts.] Find $\langle \hat{x} \rangle$, the expectation value of the position of the particle. (You should be able to guess the answer before doing the calculation.)

Solution $\rightarrow$

Guess: the mean value of the position should be at the center of the allowed region $(0, L)$, i.e., we expect $\langle \hat{x} \rangle = L/2$.

$$\begin{aligned} \langle \psi | \hat{x} | \psi \rangle &= \int_0^L dx \psi^*(x) x \psi(x) = \frac{2}{L} \int_0^L dx x \sin^2 \left(\frac{\pi x}{L} \right) \\ &= \frac{2}{L} \left(\frac{L}{\pi} \right)^2 \int_0^\pi du u \sin^2(u) \quad [\text{substitution } x = Lu/\pi] \\ &= \frac{2L}{\pi^2} \times \frac{\pi^2}{4} = \frac{L}{2} \end{aligned}$$

- (c) [7 pts.] Find the uncertainty of the position, i.e., $\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2}$.

Solution $\rightarrow$

Since the wavefunction is spread over distance L , the uncertainty should be proportional to L , but I can't guess the proportionality constant without a calculation.

$$\begin{aligned} \langle \psi | \hat{x}^2 | \psi \rangle &= \int_0^L dx \psi^*(x) x^2 \psi(x) = \frac{2}{L} \int_0^L dx x^2 \sin^2 \left(\frac{\pi x}{L} \right) \\ &= \frac{2}{L} \left(\frac{L}{\pi} \right)^3 \int_0^\pi du u^2 \sin^2(u) \quad [\text{substitution } x = Lu/\pi] \\ &= \frac{2L^2}{\pi^3} \times \left(\frac{\pi^3}{6} - \frac{\pi}{4} \right) = \left(\frac{1}{3} - \frac{1}{2\pi^2} \right) L^2 \end{aligned}$$

The uncertainty in position is

$$\begin{aligned} \Delta x &= \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} = \sqrt{\left(\frac{1}{3} - \frac{1}{2\pi^2} \right) L^2 - \left(\frac{L}{2} \right)^2} \\ &= L \sqrt{\left(\frac{1}{12} - \frac{1}{2\pi^2} \right)} \approx 0.18076L \end{aligned}$$

- (d) [6 pts.] Find the expectation value of the momentum of the particle.
(You should be able to guess the answer before doing the calculation.)

Solution $\rightarrow$

It's a stationary state, not moving in any direction. So one expects zero momentum. (This type of argument doesn't always work in quantum mechanics, but here it does.)

$$\begin{aligned}
 \langle \hat{p} \rangle &= \langle \psi | \hat{p} | \psi \rangle = \int_0^L dx \psi^*(x) \left(-i\hbar \frac{\partial}{\partial x} \right) \psi(x) \\
 &= \frac{-2i\hbar}{L} \int_0^L dx \sin\left(\frac{\pi x}{L}\right) \frac{\partial}{\partial x} \sin\left(\frac{\pi x}{L}\right) \\
 &= \frac{-2i\hbar\pi}{L^2} \int_0^L dx \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{L}\right) = \frac{-i\hbar\pi}{L^2} \int_0^L dx \sin\left(\frac{2\pi x}{L}\right) \\
 &= \frac{-i\hbar\pi}{L^2} \frac{L}{2\pi} \int_0^{2\pi} du \sin(u) \quad \left\{ \begin{array}{l} \text{[substitution } x = Lu/2\pi] \\ \text{Note different from previous substitutions} \end{array} \right. \\
 &= -\frac{i\hbar}{2L} \times 0 = 0 \quad \left\{ \begin{array}{l} \text{integral is zero because a sine function} \\ \text{is being integrated over a FULL PERIOD} \end{array} \right.
 \end{aligned}$$

Here, as always, to calculate an integral it's really worth plotting the integrand over the region of integration.

You could also do the integral using the substitution $\cos(\pi x/L) = v$, so that

$$\sin\left(\frac{\pi x}{L}\right) dx = -\frac{L}{\pi} dv.$$

However, then you have to be careful about the limits: $x = 0$ corresponds to $v = 1$, and $x = L$ corresponds to $v = -1$.

$$\langle \hat{p} \rangle = \frac{-2i\hbar\pi}{L^2} \left(-\frac{L}{\pi} \right) \int_1^{-1} v dv = -\frac{2i\hbar}{L} \int_{-1}^{+1} v dv = 0$$

- (e) [7 pts.] Find the uncertainty of the momentum.

Solution $\rightarrow$

$$\begin{aligned}
 \langle \hat{p}^2 \rangle &= \langle \psi | \hat{p}^2 | \psi \rangle = \int_0^L dx \psi^*(x) \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \psi(x) \\
 &= \frac{-2\hbar^2}{L} \int_0^L dx \sin\left(\frac{\pi x}{L}\right) \frac{\partial^2}{\partial x^2} \sin\left(\frac{\pi x}{L}\right) \\
 &= \frac{-2\hbar^2}{L} \int_0^L dx \sin\left(\frac{\pi x}{L}\right) \left[-\left(\frac{\pi}{L}\right)^2 \sin\left(\frac{\pi x}{L}\right) \right] \\
 &= \frac{2\hbar^2 \pi^2}{L^3} \int_0^L dx \sin^2\left(\frac{\pi x}{L}\right) \\
 &= \frac{2\hbar^2 \pi^2}{L^3} \left(\frac{L}{\pi}\right) \int_0^\pi du \sin^2(u) = \frac{2\hbar^2 \pi}{L^2} \times \frac{\pi}{2} = \frac{\hbar^2}{L^2} \pi^2
 \end{aligned}$$

Uncertainty of momentum

$$\Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2} = \sqrt{\frac{\hbar^2}{L^2} \pi^2 - 0^2} = \frac{\hbar \pi}{L}$$

- (f) [SELF] How does the position uncertainty depend on L ? Could you have expected this?

Solution $\rightarrow$

The particle is confined in the region $(0, L)$, hence a position measurement can give any value between 0 and L . We therefore expect the uncertainty in a position measurement to be proportional to L .

We cannot however predict the constant of proportionality from a physical argument like this; an explicit calculation is required. Indeed we have found above that Δx is a cumbersome number times L .

- (g) [**SELF**] Does the momentum uncertainty increase or decrease, if L is increased? Explain whether/how this is consistent with the Heisenberg uncertainty principle.

Solution $\rightarrow$

The uncertainty principle gives

$$\Delta x \Delta p \gtrsim \hbar$$

so it's reasonable to assume

$$\Delta p \sim \frac{\hbar}{\Delta x}$$

Since the uncertainty in position, Δx , is proportional to L , the uncertainty in momentum should be inversely proportional to L , which is what we have found above.

- (h) [**SELF**] If the energy (Hamiltonian) operator is $\hat{H} = \frac{\hat{p}^2}{2m}$, find the expectation value and the uncertainty of the energy.

Solution $\rightarrow$

$$\langle \hat{H} \rangle = \left\langle \frac{\hat{p}^2}{2m} \right\rangle = \frac{1}{2m} \langle \hat{p}^2 \rangle = \frac{1}{2m} \times \frac{\hbar^2}{L^2} \pi^2 = \frac{\hbar^2 \pi^2}{2mL^2}$$

We used here the value of $\langle \hat{p}^2 \rangle$ calculated previously.

Since the wavefunction is known to be an energy eigenfunction, the uncertainty of energy is expected to be zero. It can be calculated as follows:

$$\Delta H = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \sqrt{\frac{1}{(2m)^2} \langle \hat{p}^4 \rangle - \left(\frac{\hbar^2 \pi^2}{2mL^2} \right)^2}$$

One has to now calculate $\langle \hat{p}^4 \rangle$, which is left as an exercise.

2. Recall for a spin-1/2 system:

$$J_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad J_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad J_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Each of these matrices have two eigenvalues, in each case $+\hbar/2$ and $-\hbar/2$.

- (a) [**SELF**] Why does it make sense physically that the three operators have the same set of eigenvalues? (What's the connection between measurement results and eigenvalues?)

Solution $\rightarrow$

The choice of which direction to call x , y or z is arbitrary. Thus. the three components should have the same possible values in physical measurement. Hence the three operators should have the same set of eigenvalues.

- (b) [**3 pts.**] For J_x , verify the statement about eigenvalues above i.e., calculate eigenvalues of the J_x matrix.

Solution $\rightarrow$

Eigenvalues of $J_x = \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix}$ are the solutions of the determinant equation

$$\begin{aligned} \left| \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| &= 0 \quad \implies \quad \begin{vmatrix} -\lambda & \hbar/2 \\ \hbar/2 & -\lambda \end{vmatrix} = 0 \\ \implies \quad (-\lambda)^2 - \left(\frac{\hbar}{2}\right)^2 &= 0 \quad \implies \quad \lambda^2 = \left(\frac{\hbar}{2}\right)^2 \end{aligned}$$

The eigenvalues are therefore $+\frac{\hbar}{2}$ and $-\frac{\hbar}{2}$.

If you are not 100% comfortable with calculating eigenvalues fast, please review the technique asap. For practice, calculate the eigenvalues of the matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

- (c) [2 pts.] The eigenvector of J_x corresponding to the eigenvalue $+\hbar/2$ is $|x, +\rangle = \begin{pmatrix} \alpha \\ \alpha \end{pmatrix}$. Calculate α so that the state is normalized. You can choose α to be real and positive.

Solution $\rightarrow$

$$2|\alpha|^2 = 1 \quad \implies \quad \alpha = \frac{1}{\sqrt{2}}$$

Taking α to be real and positive.

You should of course check that $|x, +\rangle$ is indeed an eigenvector of J_x as claimed, by calculating the matrix-vector product $J_x |x, +\rangle$.

- (d) [4 pts.] Calculate the expectation value of J_x in the state $|x, +\rangle$. (You should be able to guess the answer before doing the calculation.)

Solution $\rightarrow$

$$\begin{aligned} \langle J_x \rangle &= \langle x, + | J_x | x, + \rangle = \begin{pmatrix} \alpha^* & \alpha^* \end{pmatrix} \begin{pmatrix} 0 & \hbar/2 \\ \hbar/2 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} \\ &= \begin{pmatrix} \alpha^* & \alpha^* \end{pmatrix} \begin{pmatrix} \alpha\hbar/2 \\ \alpha\hbar/2 \end{pmatrix} = 2|\alpha|^2 \frac{\hbar}{2} = \frac{\hbar}{2} \end{aligned}$$

Since $|x, +\rangle$ is an eigenstate of J_x , a measurement of this observable will give the corresponding eigenvalue, i.e., we are sure to get the result $+\hbar/2$ in a measurement of the x -component of spin. So, the expectation value of the J_x operator should be $+\hbar/2$, as we found, and the uncertainty should be zero (next problem).

- (e) [4 pts.] Calculate the uncertainty of J_x in the state $|x, +\rangle$. (You should be able to guess the answer before doing the calculation.)

Solution $\rightarrow$

We need to calculate $\sqrt{\langle J_x^2 \rangle - \langle J_x \rangle^2}$.

Note that the operator (or matrix) J_x^2 is obtained by a matrix multiplication.

$$J_x^2 = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Please note that this is NOT the same as squaring each element of J_x , i.e.,

$$J_x^2 \neq \begin{pmatrix} 0 & \hbar^2/4 \\ \hbar^2/4 & 0 \end{pmatrix}$$

Therefore

$$\begin{aligned} \langle J_x^2 \rangle &= \langle x, + | J_x^2 | x, + \rangle = \frac{\hbar^2}{4} (\alpha^* \quad \alpha^*) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} \\ &= \frac{\hbar^2}{4} (\alpha^* \quad \alpha^*) \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} = \frac{\hbar^2}{4} 2|\alpha|^2 = \frac{\hbar^2}{4} \end{aligned}$$

Thus the uncertainty is

$$\sqrt{\langle J_x^2 \rangle - \langle J_x \rangle^2} = \sqrt{\frac{\hbar^2}{4} - \left(\frac{\hbar}{2}\right)^2} = 0$$

as expected.

- (f) [4 pts.] Calculate the expectation value of J_z in the state $|x, +\rangle$. (You might be able to guess the answer before doing the calculation.)

Solution $\rightarrow$

What do we expect? $|x, +\rangle$ is an eigenstate of J_x , not of J_z . It's reasonable to assume that the $|x, +\rangle$ state does not prefer either a $+\hbar/2$ or a $-\hbar/2$ as the result for a measurement of J_z . (These are the only two outcomes of a J_z measurement.) So, one expects equal probabilities for these two possible outcomes, if J_z is measured in the $|x, +\rangle$ state. One can thus expect the expectation value of J_z to be the average of these two values, i.e., 0. Let's see if this works:

$$\begin{aligned}\langle J_z \rangle &= \langle x, + | J_z | x, + \rangle = \begin{pmatrix} \alpha^* & \alpha^* \end{pmatrix} \begin{pmatrix} \hbar/2 & 0 \\ 0 & -\hbar/2 \end{pmatrix} \begin{pmatrix} \alpha \\ \alpha \end{pmatrix} \\ &= \begin{pmatrix} \alpha^* & \alpha^* \end{pmatrix} \begin{pmatrix} \alpha\hbar/2 \\ -\alpha\hbar/2 \end{pmatrix} = |\alpha|^2 \frac{\hbar}{2} - |\alpha|^2 \frac{\hbar}{2} = 0\end{aligned}$$

- (g) [4 pts.] Calculate the uncertainty of J_z in the state $|x, +\rangle$.

Solution $\rightarrow$

One expects this to be nonzero, since outcomes $+\hbar/2$ and $-\hbar/2$ are equally probable. I can't however predict the exact value without doing the calculation.

Note that $J_z^2 = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = J_x^2$. Thus

$$\langle J_z^2 \rangle = \langle J_x^2 \rangle = \frac{\hbar^2}{4}$$

The uncertainty is

$$\Delta J_z = \sqrt{\langle J_z^2 \rangle - \langle J_z \rangle^2} = \sqrt{\frac{\hbar^2}{4} - 0^2} = \frac{\hbar}{2}$$

3. A single particle on a one-dimensional line is in the state described by the wavefunction

$$\psi(x) = B \exp \left[-\frac{(x-a)^2}{\sigma^2} \right]$$

- (a) **[SLEF]** Calculate the constant B so that $\psi(x)$ is normalized.
- (b) **[SELF]** Find the expectation value of the position of the particle. (You should be able to guess the answer before doing the calculation. Also, you should be able to do the definite integral by symmetry.)
- (c) **[SELF]** Find the uncertainty of the position. (You should be able to guess the answer within a factor of two, before doing the calculation.)
- (d) **[SELF]** Find the expectation value of the momentum of the particle.
- (e) **[SELF]** Find the uncertainty of the momentum. How is the uncertainty of the momentum related to the uncertainty of the position?

4. The wavefunction of a particle on a one-dimensional line is given by

$$\psi(x, t) = C(t) \exp \left[-\frac{(x - vt)^2}{(s_0 + \xi t)^2} \right]$$

Expectation values (and uncertainties) are functions of time, since the wavefunction is time-dependent.

You should be able to answer the following based on the previous problem, without new lengthy calculations.

- (a) **[SELF]** Calculate the time-dependent (but space-independent) quantity $C(t)$ so that the wavefunction is normalized at all times.
- (b) **[SELF]** Express and plot $\langle x(t) \rangle$ as a function of time.
- (c) **[SELF]** Plot the uncertainty of the position, as a function of time.
- (d) **[SELF]** Plot the uncertainty of the momentum, as a function of time.